Time series analysis — exercise sheet 1

AB–204 Arctic population ecology, October 27, 2026

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This exercise sheet is about working with data in R and visualising (time series) data in R (Lecture 1).

Before you start: save `bike_sharing.csv` (from the course website) in your course folder `AB-204`, open a new R script (File → New File → R Script), and load the colours for categories from the slides at the top of the script:

```
library(viridisLite) # viridis(n, end = 0.9): n colours, one per category
```

Exercise 1: working with data in R

1. Use the `setwd()` function to set your course folder as working directory.
2. Use the `read.csv()` function to import the file `bike_sharing.csv`, and the assignment operator `<-` to assign it to the object `bikes`.
3. Get an overview of the data. How many variables are there? How many observations? Are there any missing values — and if so, in which variables? What are the scales of measurement?
   Hint: `dim()`, `head()`, `str()`, `summary()`.
4. In the data, the wind speed is given in miles per hour, and the temperature in degrees Fahrenheit. Create the new variables `windspeed_kmh` and `temperature_c` that give the wind speed in kilometres per hour and the **minimum** temperature in degrees Celsius, respectively. Use

$$\text{kilometres per hour} = \text{miles per hour} \times 1.609,$$
$$\text{degrees Celsius} = (\text{degrees Fahrenheit} - 32) \times 5/9.$$

   Also convert the variable `date` to a date using `as.Date()`.
5. Create a subset of the data for 2025 and assign it to the object `bikes_2025`. Use this subset for Exercises 2 and 3.

Exercise 2: visualising time series and relationships in R

1. Create a time series plot of `count` in 2025. What patterns can you identify? Is there anything unusual?
2. Use the `plot()` function to create pairwise scatter plots of the different variables against `count`. Based on a visual inspection of the scatter plots, which variables seem to affect the number of bikes most?
3. Use the `simply_scatter()` function from the `simply3d` package to create a 3-dimensional scatter plot of `windspeed_kmh` against `temperature_c` against `count`.
4. Use different colours for rainy days (`precipitation > 0`) and dry days in your 3-dimensional scatter plot from 2.3. Can you see a difference between rainy and dry days?

Exercise 3: visualising distributions in R

1. Use the `boxplot()` function to create a boxplot of `count`. Based on the boxplot, what can you say about the distribution?
2. Use the `hist()` function to create a histogram of `count`. Try different numbers of bins (argument `breaks`).
3. Use the `density()` function to add a kernel density estimate to the histogram from 3.2.
   Hint: use `probability = TRUE` in `hist()`, and `lines()` to add the density.
4. Visualise the distribution of `count` separately for rainy and dry days. What do you see?
5. **Bonus:** Visualise the distribution of `count` for each year (full data set). How does it change?